

Import Price Shocks, FX-Monetary Policies, and Real Income Stabilization

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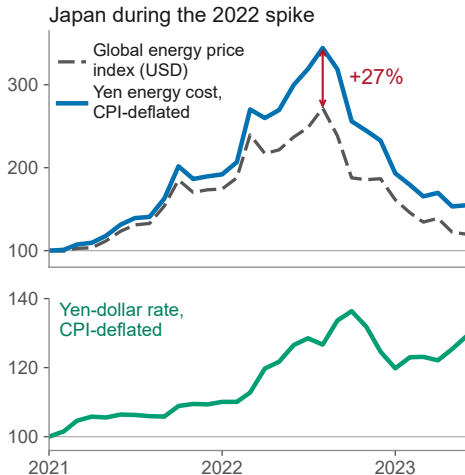
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Motivation: Essential Import Price Spikes

- Recent years: **import-dependent economies** facing recurrent episodes of large, temporary spikes in essential-import prices:
 - 2022: LNG and crude +100–500%
 - 2026 Q1: crude +60%; Asian LNG +140%
- A **negative terms-of-trade** shock: give away more domestic output to buy the import. Price-inelastic demand: translates into **real-income losses**.

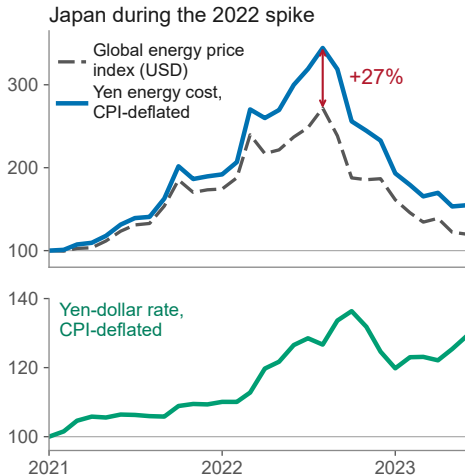
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- In the data: concurrent **real depreciation** endogenously worsens the terms of trade and amplifies real-income losses.
- Policy concerns in a monetary economy: **imported inflation** and nominal depreciation.



How should monetary and exchange rate policies respond?

Existing Policy Views

"It is far simpler to allow one price to change, namely, the price of foreign exchange. . . "

Friedman (1953), p. 173

- **Textbook open-economy RA NK framework:**
 - **"Raise rates"**: use monetary policy to anchor inflation; contractionary stance that limits aggregate demand and domestic activity.
 - **Exchange rate as a shock absorber**: do not manage the real exchange rate. **Let it depreciate.**
Rationale: depreciation reflects expenditure switching. With perfect asset-market access, a higher import bill only a PDV loss: households optimally smooth over time.
- **ToT shocks in HANK**: laissez-faire exchange rates. **Real-income channel**: monetary policy less contractionary than RA; activity expansion indirectly supports constrained households' real incomes.
- Typically: monetary policy for fighting inflation and **protecting real incomes** of constrained households; FX policy for managing volatile capital flows, financial crises, and market-power motives.
- The policy set: **monetary policy** operates through nominal rigidities and influences domestic activity; **FX policy** influences the real exchange rate path by managing capital flows.

This Paper

- Open-economy framework with **heterogeneous households**: some borrowing constrained.
- **Marshall-Lerner condition**: inelastic import demand. Downward-sloping, price-elastic export demand: **real depreciation** required for higher export revenues.
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How does heterogeneity change the roles of FX and monetary policies? To what extent can they protect households who cannot borrow through the shock?

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- A passive laissez-faire “let it depreciate” exchange rate response costly for constrained households.
 - Instead, should **lean against a contemporaneous depreciation** to **protect real incomes**. **Active role for FX policy in fighting ToT shocks and imported inflation**.
 - **The real-income stabilization channel**: induces stronger capital inflows during a spike and relies on the Marshall-Lerner condition to appreciate the RER and **indirectly** boost real incomes.
 - **FX policy as anti-inflation policy**: more effective than monetary policy in protecting real incomes of constrained households and fighting imported inflation. Drives **interest parity wedges**.

Results and Contributions

- **Normative analysis.** Two-agent model. Optimal FX and monetary policy response to a ToT shock.
 - **FX policy:** isolate heterogeneity-induced role. **Real-income stabilization: indirect income effects** that rely on the Marshall-Lerner condition. Role increasing in essentiality of imports.
 - **Monetary policy:** supports FX policy under employment-import complementarity. **More contractionary than RA: redistributive price effects**; import-induced depreciations redistribute from constrained to unconstrained.
- **Quantitative two-agent model.** Nominal wage rigidity, financial intermediation frictions. Utilitarian-optimal FX interventions and monetary policy under perfect-foresight transitions. Application: **energy-price shocks**. Japan calibration, 2022-like price path.
- **FX policy leads with real-income protection and disinflation. MP alone does little:**
 - Joint optimum vs. conventional: energy **19.49%** cheaper; **29.03%** of real wage loss undone. Average inflation: **0.60% → 0.23%**.
 - Monetary policy alone: **0.27%** of wage loss undone; average inflation: **0.60% → 0.58%**.
- **HA framework with idiosyncratic risk:** resolve a distributional time inconsistency. Optimal nonlinear policies under timeless commitment. Computational: exploit affine form of the promise-keeping state.

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- **Conventional open-economy policy.** Use contractionary monetary policy to fight inflation and let the exchange rate adjust freely with undistorted capital flows.
Friedman (1953); Galí and Monacelli (2005)
- **HA real-income transmission in closed and open economies.** Monetary policy reaches constrained households through activity-induced indirect real-income effects.
Kaplan, Moll, and Violante (2018); Auclert (2019); Auclert et al. (2024)
- **Intermediation frictions and FX interventions.** Study exchange rates and FX policy for closing intermediation frictions and stabilizing exchange rates under capital-flow volatility.
Gabaix and Maggiori (2015); Fanelli and Straub (2021); Itskhoki and Mukhin (2023)
- **Energy shocks and HANK.** Monetary transmission and household incidence under energy shocks.
Blanchard and Galí (2007); Auclert et al. (2023); Chan, Diz, and Kanngiesser (2024)
- **Optimal policy in HANK.** Monetary policy redistributes; transfers target.
Acharya, Challe, and Dogra (2023); Dávila and Schaab (2023); McKay and Wolf (2026)

A Minimal Analytical Framework

- Two-period, two-agent framework. Real model, no nominal rigidity, no monetary policy. **Goal: isolate the role of FX policy under heterogeneity.**
- Open economy: small in asset and import markets; large in export market.
- **Domestic output** Y_t : produced using labor and an imported input. Output is tradable: consumed at home or exported abroad:

$$C_t + X_t = Y_t$$

- **Foreign good (dollars/consumption basket)**: not produced or consumed at home; world numeraire.
- **Imported input**: price P_t^{T*} in foreign units.
- **Asset markets**: economy accesses international asset markets; world assets **denominated in foreign good units**; world real interest rate R^* .
- Economy consumes domestic good only. P_t : nominal price and price level; e_t : nominal exchange rate. **Real exchange rate**: $\epsilon_t = e_t/P_t$, relative price of foreign good in domestic units.
- **Terms of trade**: $\epsilon_t P_t^{T*}$, relative price of the imported input in domestic units.

Firms and Production Technology

- Domestic output is produced with labor and the imported input using a Leontief technology:

$$Y_t = A \cdot \min \left\{ N_t, \frac{M_t}{\theta} \right\}$$

- Cost function: $C_t = w_t N_t + \epsilon_t P_t^T * M_t$
- Cost minimization implies $M_t = \theta N_t$, $Y_t = A N_t$. Marginal cost: $\mathcal{MC}_t = w_t + \epsilon_t P_t^T * \theta$
- Zero-profit condition: $Y_t = w_t N_t + \epsilon_t P_t^T * \theta N_t$, which gives:

$$w_t = A - \epsilon_t P_t^T * \theta$$

- A unit of labor with θ units of the imported input produces A units of output, of which w_t accrues to the worker and $\epsilon_t P_t^T * \theta$ is paid to buy the imported input.
- A higher import price or a real depreciation lowers the real wage one-for-one to restore zero profits in equilibrium.

Export Demand

- Downward-sloping demand for exports. World demand for the domestic good:

$$X_t = \zeta \epsilon_t^{1+\chi}$$

- $(1 + \chi)$ is the price elasticity of export demand, with $\chi > 0$: relatively price-elastic demand.
- Export revenue in foreign units:

$$\frac{X_t}{\epsilon_t} = \zeta \epsilon_t^{\chi}$$

- **Marshall-Lerner condition:** $0 < \chi < \infty$, a real depreciation required to generate higher foreign purchasing power.

Households

- Two household types: **hand-to-mouth** (mass $\lambda \in [0, 1]$) and **Ricardian** (mass $1 - \lambda$).
- Consumption: C_t^P, C_t^R . Ricardians participate in asset markets. Asset position: f^* .
- Both types supply N units of labor inelastically.
- FX policy**: capital control tax τ on Ricardian positions, rebated lump-sum T .
- No fiscal redistribution: HtMs receive no direct transfers.

Ricardians (mass $1 - \lambda$)

$$\max_{C_t^R, f^*} u(C_1^R) + \beta u(C_2^R) \quad \text{s.t.}$$

$$C_1^R + (1 + \tau) \epsilon_1 f^* = w_1 N + T$$

$$C_2^R = w_2 N + \epsilon_2 R^* f^*$$

Euler equation:

$$\underbrace{\frac{u'(C_1^R)}{\beta u'(C_2^R)}}_{\text{Domestic real rate}} = \underbrace{\frac{1}{(1 + \tau)}}_{\text{Wedge}} \underbrace{R^* \frac{\epsilon_2}{\epsilon_1}}_{\text{Effective world real rate}}$$

Hand-to-Mouth (mass λ)

- No asset market access.
- Consume current incomes each period:

$$C_t^P = w_t N$$

Balance of Payments

- Period-by-period external budget constraints:

$$P_1^T \theta N + F^* = \zeta \epsilon_1^X$$

$$P_2^T \theta N = \zeta \epsilon_2^X + R^* F^* \quad F^* = \frac{f^*}{1 - \lambda}$$

- BoP identity: trade balance = net capital outflows $\equiv NX_t$:

$$\underbrace{NX_t}_{\text{Trade balance}} = \underbrace{\zeta \epsilon_t^X}_{\text{Export revenue}} - \underbrace{P_t^T \theta N}_{\text{Import bill}}$$

- Aggregate intertemporal budget constraint:

$$\underbrace{P_1^T \theta N + \frac{1}{R^*} P_2^T \theta N}_{\text{PDV of import bills}} = \underbrace{\zeta \epsilon_1^X + \frac{1}{R^*} \zeta \epsilon_2^X}_{\text{PDV of export revenues}}$$

- Leontief + inelastic labor supply = perfectly inelastic import demand.

A Utilitarian Ramsey Planner's Problem

- Maximize population-weighted social welfare:

$$\max_{\{\epsilon_t\}} \lambda [u(C_1^P) + \beta u(C_2^P)] + (1 - \lambda) [u(C_1^r) + \beta u(C_2^r)] \quad \text{s.t.}$$

$$NX_1(\epsilon_1) + \frac{1}{R^*} NX_2(\epsilon_2) = 0$$

where

$$Y_t^N \equiv (A - \epsilon_t P_t^{T*} \theta) N \quad \text{real labor income}$$

$$C_t^P = Y_t^N$$

$$C_t^r = Y_t^N - \frac{\epsilon_t NX_t}{1 - \lambda}$$

- Primal formulation: Ricardian Euler equation not an implementability constraint.
- Lemma.** The solution to this problem is implementable as a decentralized equilibrium with an appropriate FX policy (τ, T) .

Optimal Policy

- Assume $\beta R^* = 1$. Laissez-faire equilibrium: $\tau = 0$.

Proposition: Representative agent ($\lambda = 0$)

For any $(P_1^{T*}, P_2^{T*}) > 0$, the optimal allocation involves **no intertemporal distortions**: $\tau = 0$.

- All agents smooth consumption perfectly given any price sequence.
- Special case: no ToT manipulation to internalize export market power.

Proposition: Heterogeneous agents ($\lambda \in (0, 1)$)

If $P_1^{T*} = P_2^{T*}$, then $\tau = 0$ for any λ .

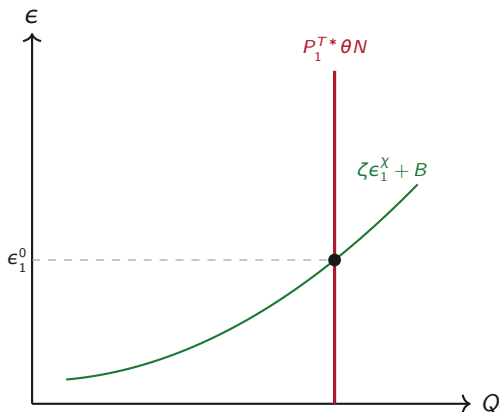
If $P_1^{T*} > P_2^{T*}$, with $(\epsilon_t^{lf}, Y_t^{N,lf})$ laissez-faire and $(\epsilon_t^O, Y_t^{N,O}, \tau^O)$ the constrained efficient allocation:

- Weaker RER co-movement:** appreciation in response to the shock: $\epsilon_1^O < \epsilon_1^{lf}$, $\epsilon_2^O > \epsilon_2^{lf}$
- Real-income and ToT stabilization:** $|Y_2^{N,O} - Y_1^{N,O}| < |Y_2^{N,lf} - Y_1^{N,lf}|$, $|\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*}| < |\epsilon_1^{lf} P_1^{T*} - \epsilon_2^{lf} P_2^{T*}|$
- More volatile capital flows:** $NX_1^O < NX_1^{lf} < 0 < NX_2^{lf} < NX_2^O$
- Strictly positive intertemporal wedge:** $\tau^O > 0$

What Happened?

- Subsidize Ricardians to induce **stronger net capital inflows** in the high-price period. Import demand perfectly inelastic: capital inflows substitute for export revenues and **appreciate** the real exchange rate.
- Fewer exports and more resources retained at home at $t = 1$.
- At $t = 2$, after the spike has passed, must tolerate more depreciation: capital outflows and exports.
- Weaker export and RER co-movement with the shock. Stronger capital-inflow co-movement.
- **Why is this optimal? Smoother consumption path for HtM households through the shock.**
- No direct fiscal transfers: any additional policy-induced borrowing accrues to Ricardians and distorts their consumption path. **How then does this policy smooth HtM consumption?**
- **The real-income stabilization channel:** capital-flow-induced RER appreciation raises real wages for **all** workers and **indirectly** boosts their purchasing power. Period 2 reverses this.
- **Marshall-Lerner condition** is the key: without it, policy cannot appreciate and boost real incomes.

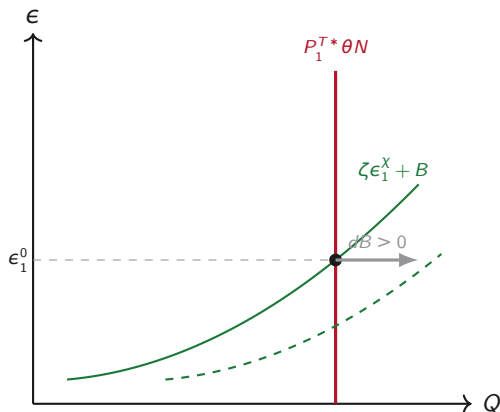
The Foreign Exchange Market and Indirect Income Effects



- BoP condition as the foreign exchange market: equating demand and supply of **foreign good**:

$$\underbrace{P_1^T * \theta N}_{\text{Demand}} = \underbrace{\zeta\epsilon_1^X + B}_{\text{Supply}}$$

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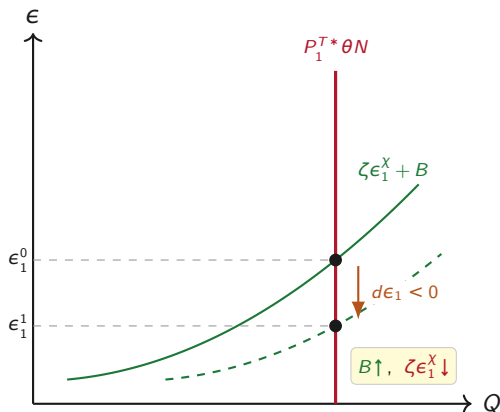


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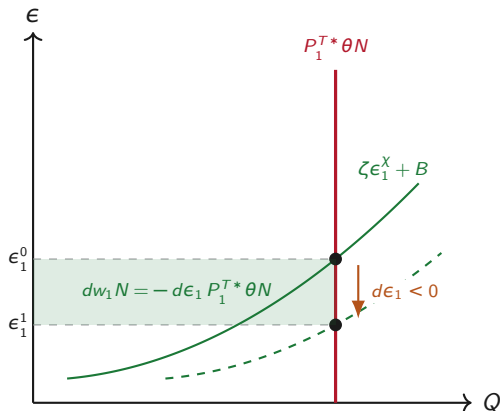


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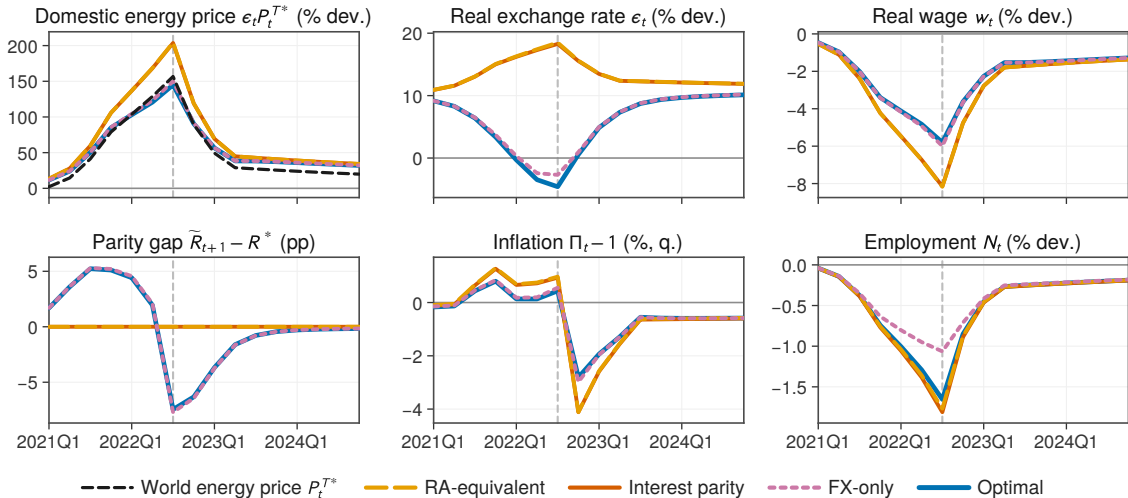
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- FX policy**: Ricardians borrow more, $dB > 0$, and sell the foreign good.
- Ricardians bid down the RER. Foreigners **crowded out**: export revenues fall.
- Cheaper input, firms compete for workers and **bid the real wage up** to restore zero profits: $w_1 = A - \epsilon_1 P_1^T * \theta$. **Every worker gains**.
- Marshall-Lerner elasticity crucial: effect disappears as $\chi \rightarrow \infty$. Stronger as $\chi \rightarrow 0$.

Taking Stock: FX Policy and Real Income Stabilization

- Borrowing constraints and the Marshall-Lerner condition turn the exchange rate from a **shock absorber into a shock amplifier**: during an essential-import price spike, constrained households face amplified current real-income losses due to a concurrent RER depreciation.
- **The real-income stabilization channel of exchange rate policy**: induce stronger capital inflows than laissez-faire to lean against the depreciation. Real wages rise for all workers: gains reach constrained households through indirect income effects.
- Policy re-times the PDV loss but does not eliminate it: tolerates more depreciation, higher exports, and lower real wages after the spike. Achieves a smoother consumption path for constrained households.
- Works by driving **interest parity wedges**. Optimal real rate endogenous. Implementable through FXI.
- Requires no nominal rigidities; works independently of real activity levels: output and employment.

Quantitative Results Sneak Peek: 2022 Spike



Appendix

Elastic Labor Supply and Monetary Policy

- Both types now supply N_t units of labor, with disutility $v(N_t)$; households do not choose N_t .
- Rigid nominal wage** $\overline{W}$. Phillips curve: $P_t = \overline{W}/w_t$.
- Monetary policy** sets the money supply, and the exchange equation determines nominal demand:

$$\mathcal{M}_t = P_t Y_t = P_t A N_t$$

- Monetary policy and the Phillips curve together determine the price level and employment. Real wage determined by firms' ZPC as before.

The planner's problem revisited:

$$\max_{\{N_t, \epsilon_t\}} \lambda [u(C_1^p) + \beta u(C_2^p)] + (1 - \lambda) [u(C_1^r) + \beta u(C_2^r)] - [v(N_1) + \beta v(N_2)] \quad \text{s.t.}$$

$$NX_1(\epsilon_1, N_1) + \frac{1}{R^*} NX_2(\epsilon_2, N_2) = 0$$

Lemma. The solution is implementable as a decentralized equilibrium with fiscal-monetary policy $(\tau, T, \mathcal{M}_t)$.

Optimal Policy: Elastic Labor Supply

Proposition: Representative agent ($\lambda = 0$)

For any $(P_1^{T*}, P_2^{T*}) > 0$, the optimal allocation involves no intertemporal distortions: $\tau = 0$. The labor wedge is distorted to account for the **static terms-of-trade correction**:

$$v'(N_t) = u'(C_t) \left(w_t - \frac{\epsilon_t P_t^{T*} \theta}{\chi} \right)$$

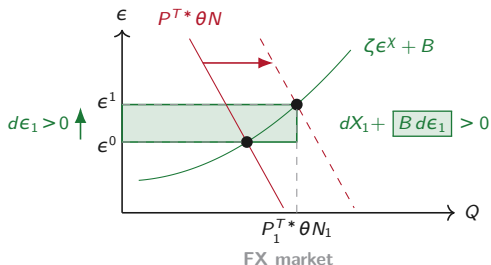
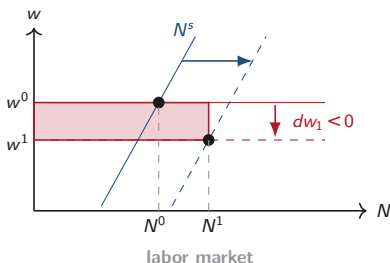
Proposition: Heterogeneous agents ($\lambda \in (0, 1)$)

- $P_1^{T*} = P_2^{T*}$: no intertemporal distortions; labor wedge corrects for the terms-of-trade effect.
- $P_1^{T*} > P_2^{T*}$: positive intertemporal wedge, $\tau > 0$; labor wedge corrects for both the static terms-of-trade effect and a **redistributive price effect**:

$$v'(N_t) = \bar{u}'_t \left(w_t - \frac{\epsilon_t P_t^{T*} \theta}{\chi} \right) + \lambda [u'(C_t^p) - u'(C_t^r)] NX_t \frac{d\epsilon_t}{dN_t} \Big|_{NX_t}$$

A Redistributive Price Effect

- Suppose $P_1^{T*} > P_2^{T*}$. Holding $NX = -B < 0$ fixed, consider a marginal increase in labor supply.



- $dN > 0 \Rightarrow$ workers bid down wage $\Rightarrow$ firms expand: import demand $\uparrow \Rightarrow \epsilon_1 \uparrow, w_1 \downarrow$
- Workers gain from more employment, but lose from a lower wage. Net gains are asymmetric: **all borrowers are workers but not all workers are borrowers.**
- Ricardian borrowers receive a pecuniary benefit from this depreciation: $\epsilon B \uparrow$. Real wages lost by workers finance this gain: a net endogenous **transfer from HtMs to Ricardians.**

Taking Stock: Joint Optimal Policy

- Relative to RA: during spike episodes, **FX policy induces stronger capital inflows** and **monetary policy is more contractionary**.
- FX policy distorts the **intertemporal wedge**. Monetary policy distorts the static **labor wedge** within each period and, under employment-import complementarity, influences import demand.
- Both influence the FX market through different channels and thereby influence the equilibrium RER.
- FX policy seeks appreciation and higher real wages through capital inflows. Realized appreciation depends on how much monetary policy allows employment to expand.
- Feasible: FX policy induces capital inflows and monetary policy allows higher employment and imports, but must tolerate more depreciation and lower real wages. ToT externality hurts workers; redistributive price effects hurt constrained households.
- Also feasible: monetary policy restrains employment. FX-policy-induced capital inflows appreciate the RER and raise real wages: lower prices under rigid nominal wages.
- **Optimal**: real-income gains to constrained through **higher wages, restrained employment**.

Monetary Policy in Isolation

- Without FX policy, monetary policy faces the Ricardian Euler equation, i.e., the **interest parity condition** as an implementability constraint:

$$\frac{u'(C_1^r)}{\beta u'(C_2^r)} = R^* \frac{\epsilon_2}{\epsilon_1}$$

- Ricardian budget constraints and the Euler equation imply the Ricardian **saving function**:

$$F^* = F^*(\epsilon_1, \epsilon_2, N_1, N_2)$$

- Monetary policy sets the employment path, $\{N_1, N_2\}$; capital flows and the RER path follow as a consequence of balance of payments:

$$\begin{aligned} P_1^T \theta N_1 + F^*(\epsilon_1, \epsilon_2, N_1, N_2) &= \zeta \epsilon_1^X \\ P_2^T \theta N_2 &= \zeta \epsilon_2^X + R^* F^*(\epsilon_1, \epsilon_2, N_1, N_2) \end{aligned}$$

- FX policy breaks this dependency, moving F^* independently of the employment path through interest parity wedges: tighter grip over the RER, independent of the level of real activity.

Quantitative Model

- Baseline: two-agent infinite-horizon framework.
- Segmented asset markets: Ricardians cannot access international assets directly. Domestic assets: nominal units; international assets: foreign-good units.
- Ricardians solve:

$$\max_{\{C_t^r, B_{t+1}^{M,r}\}} \sum_{t=1}^{\infty} \beta^{t-1} [u(C_t^r) - v(N_t)] \quad \text{s.t.}$$
$$P_t C_t^r + B_{t+1}^{M,r} = W_t N_t + R_t^M B_t^{M,r} + T_t$$

- Optimality:

$$\frac{u'(C_t^r)}{\beta u'(C_{t+1}^r)} = R_{t+1}, \quad \lim_{T \rightarrow \infty} \frac{B_{T+1}^r}{\prod_{s=2}^T R_s} = 0, \quad B_{t+1}^r = \frac{B_{t+1}^{M,r}}{P_t}$$

- $R_{t+1} = \frac{R_{t+1}^M}{\Pi_{t+1}}$: real interest rate in domestic units; $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$ gross inflation rate.

Nominal Wage Rigidity and the Labor Wedge

- Adapted from Erceg, Henderson, and Levin (2000) and Auclert, Rognlie, and Straub (2024).
- Final labor service produced by an aggregator using $k \in [0, 1]$ tasks.
- For each task k , a union sets the nominal wage $W_t(k)$ for a continuum of workers.
- Each union faces a **private utility cost** of changing the nominal wage.
- Wage Phillips curve with symmetric unions:

$$v'(N_t) = (1 - \tau_t^N) w_t \bar{u}'_t,$$
$$1 - \tau_t^N = \frac{1}{\mu_w} + \frac{\left(\frac{w_t}{w_{t-1}} \Pi_t - 1 \right) \frac{w_t}{w_{t-1}} \Pi_t - \beta \left(\frac{w_{t+1}}{w_t} \Pi_{t+1} - 1 \right) \frac{w_{t+1}}{w_t} \Pi_{t+1}}{\frac{\varepsilon_w}{\psi_{nr}} w_t N_t \bar{u}'_t}$$

International Financial Intermediaries

- Adapted from Gabaix and Maggiori (2015).
- Take positions in domestic and international asset markets s.t. a balance sheet constraint:
$$\frac{F_{t+1}}{e_t} + F_{t+1}^* = 0.$$
- Intermediary returns:

$$V_{t+1} = \left[R^* F_{t+1}^* + R_{t+1}^M \frac{F_{t+1}}{e_{t+1}} \right] = \left[R^* - R_{t+1}^M \left(\frac{e_t}{e_{t+1}} \right) \right] F_{t+1}^*$$

- Seek to maximize returns subject to a limited commitment friction:

$$\max \frac{1}{R^*} V_{t+1} \quad \text{s.t.} \quad \frac{1}{R^*} V_{t+1} \geq \Gamma F_{t+1}^{*2}$$

- Optimal asset position a function of the interest parity gap:

$$F_{t+1}^* = \frac{1}{\Gamma} \left[1 - \frac{R_{t+1}^M}{R^*} \left(\frac{e_t}{e_{t+1}} \right) \right] = \frac{1}{\Gamma} \left[1 - \left(\frac{R_{t+1}}{R^*} \frac{\epsilon_t}{e_{t+1}} \right) \right]$$

Fiscal-Monetary Policy and Accounting

- Monetary policy sets money growth. Nominal aggregate demand follows the exchange equation:

$$g_t^{\mathcal{M}} = \Pi_t \frac{Y_t}{Y_{t-1}}$$

Equivalently, it determines the ex post real rate R_1 and forward-looking nominal rates:

$$R_{t+1}^M = R_{t+1} \Pi_{t+1} \quad \text{for } t \geq 1$$

- Fiscal policy: domestic and foreign asset positions, with budget constraint:

$$B_{t+1}^{M,G} + e_t B_{t+1}^{G*} + (1 - \lambda) T_t = R_t^M B_t^{M,G} + e_t R^* B_t^{G*}$$

$$\lim_{T \rightarrow \infty} \frac{B_{T+1}^G}{\prod_{s=2}^T R_s} = 0, \quad \lim_{T \rightarrow \infty} \frac{B_{T+1}^{G*}}{R^{*T-1}} = 0, \quad B_{t+1}^G \equiv \frac{B_{t+1}^{M,G}}{P_t}$$

- Bond-market clearing:

$$B_{t+1} + B_{t+1}^G = \epsilon_t F_{t+1}^*$$

- Balance of payments:

$$P_t^{T*} M_t + F_{t+1}^* + B_{t+1}^{G*} = \zeta \epsilon_t^X + R^* (F_t^* + B_t^{G*}) - R^* \Gamma F_t^{*2}$$

Ramsey Problem

$$\max_{\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} [\lambda u(C_t^p) + (1-\lambda)u(C_t^r) - v(N_t)] \quad \text{s.t.}$$

$$\underbrace{\sum_{t=1}^{\infty} \frac{\frac{1}{\Gamma} \left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right)^2}{R^{*t-1}}}_{\text{carry trade cost}} + \sum_{t=1}^{\infty} \frac{P_t^{T*} \theta N_t}{R^{*t-1}} = \sum_{t=1}^{\infty} \frac{\zeta \epsilon_t^{\chi}}{R^{*t-1}} + R^* B_1^{G*} + R_1 \frac{\epsilon_0 F_1^*}{\epsilon_1}$$

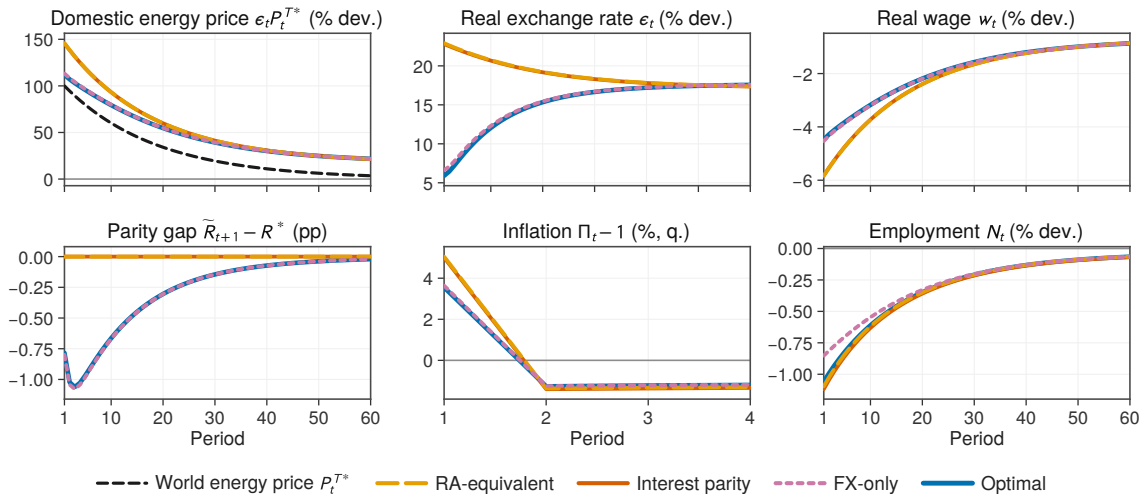
$$\tilde{R}_{t+1} = \frac{u'(C_t^r) \epsilon_t}{\beta u'(C_{t+1}^r) \epsilon_{t+1}}, \quad C_t^r = A_t N_t - \frac{\zeta \epsilon_t^{1+\chi}}{1-\lambda} + \frac{\lambda}{1-\lambda} \epsilon_t P_t^{T*} \theta N_t, \quad C_t^p = (A_t - \epsilon_t P_t^{T*} \theta) N_t$$

- Qualitatively the same primal problem as the analytical framework, except for the **carry cost** term in the IBC: interest parity wedges are costly in the form of carry trade costs from intermediary profits.
- Analytical framework nested: $\Gamma \rightarrow 0$, the same primal problem, decentralized with capital controls.
- **Lemma.** The solution to this problem is implementable as a decentralized equilibrium.
- Phillips curve not an implementability constraint: costless non-neutrality; inflation path implements the optimal labor wedge to satisfy the Phillips curve.

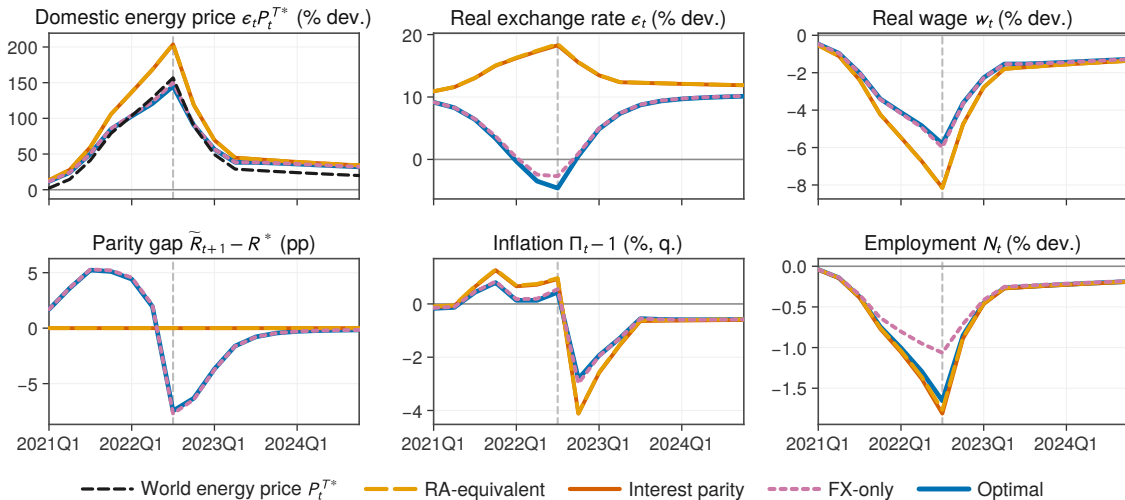
Application: Energy Price Shocks

- Essential import: fossil fuels. Calibrated to Japan: 98% import dependence; over 4% of output spent on fossil fuels in 2022.
- Compare jointly optimal policy under perfect foresight to three counterfactual policies:
 - **RA-equivalent**: no intertemporal distortions; labor wedge corrects the terms-of-trade distortion.
 - **Interest parity**: FX policy closes the intermediation friction; optimal monetary policy in isolation.
 - **FX-only**: intertemporal stance free; labor wedge corrects RA-like terms-of-trade distortion only.
- Policy response under two import-price paths:
 - **MIT shock**: perfect-foresight impulse response.
 - **2022 spike**: response to a deterministic 2022-like energy-price path.

Transitory Spike



2022 Spike



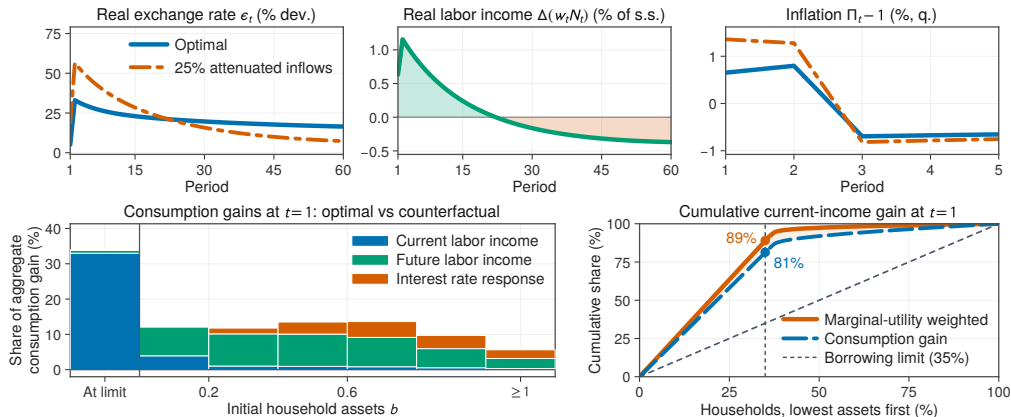
2022 Spike: Policy Comparison

2021–23 path	RA-equiv.	Interest parity	FX-only	Optimal
Peak energy-price amplification factor	1.30	1.30	0.96	0.92
Peak real-wage loss undone (%)	–	0.27	26.64	29.03
Peak domestic energy price (% lower)	–	0.18	17.88	19.49
Avg. quarterly inflation through peak (%)	0.60	0.58	0.28	0.23

- At the peak of the crisis, heterogeneity accounts for 71.06% of the contemporaneous depreciation under RA-equivalent policy relative to a world with no HtM households.
- Monetary policy set optimally, with the intertemporal stance suppressed, does little on its own to protect HtM households: it faces the interest parity condition.
- **FX policy leads with real-income protection and disinflation.**
Monetary policy supports with a disinflationary stance.
- Optimal policy re-times when the country pays for real-income losses, does not eliminate them: present value of aggregate consumption differs by only 0.0005% between optimal and RA-equivalent policies.

Heterogeneous Agents: The Income Effect across the Distribution

- Optimal FX policy vs. counterfactual: optimal policy appreciates and tilts real incomes toward high-import-price periods. Consumption response differs across the distribution.
- Current real-income gains are valued most near the borrowing limit.



Conclusion

- Borrowing constraints and the Marshall-Lerner condition turn the exchange rate from a **shock absorber into an inflation amplifier**: amplifies current real-income losses for constrained households during essential-import price spikes.
- **FX policy as an anti-inflation policy**. Managing the capital-flow path to influence the RER is more effective in addressing imported inflation and protecting real incomes than monetary policy.
- FX policy works through **indirect income effects**: it boosts households' real incomes by leaning against the depreciation during spikes and weakens the co-movement of the RER with import prices.
- Monetary policy in isolation is limited by the interest parity restriction: it cannot drive intertemporal wedges. Most real-income protection lives on this margin.

Equilibrium Definition

Given an exogenous sequence $\{P_t^{T*}\}$ and exchange rate policy (τ, T) , an equilibrium consists of $\{C_t^P, C_t^r, F^*, w_t, \epsilon_t\}$ such that:

- HtM households consume current income: $C_t^P = w_t N$
- Ricardian households optimize: $u'(C_1^r)\epsilon_1(1 + \tau) = \beta R^* u'(C_2^r)\epsilon_2$
- Firms optimize: $w_t = A - \epsilon_t P_t^{T*} \theta$
- Fiscal budget holds: $T = \tau \epsilon_1 \frac{F^*}{1 - \lambda}$
- Domestic goods market clears:

$$\lambda C_t^P + (1 - \lambda) C_t^r + \zeta \epsilon_t^{1+\chi} = AN$$

- Balance of payments:

$$P_1^{T*} \theta N + F^* = \zeta \epsilon_1^\chi, \quad P_2^{T*} \theta N = \zeta \epsilon_2^\chi + R^* F^*$$

Optimality Conditions

Optimal intertemporal wedge:

$$\epsilon_1 u'(C_1^r) + \Omega_1 = \beta R^* [\epsilon_2 u'(C_2^r) + \Omega_2]$$

$$\Omega_t \equiv \frac{\lambda [u'(C_t^p) - u'(C_t^r)] \left(\frac{\epsilon_t}{\chi} - NX_t \frac{d\epsilon_t}{dNX_t} \right)}{\frac{1+\chi}{\chi}}$$

$$\tau = \frac{\Omega_1 - \beta R^* \Omega_2}{\epsilon_1 u'(C_1^r)}$$

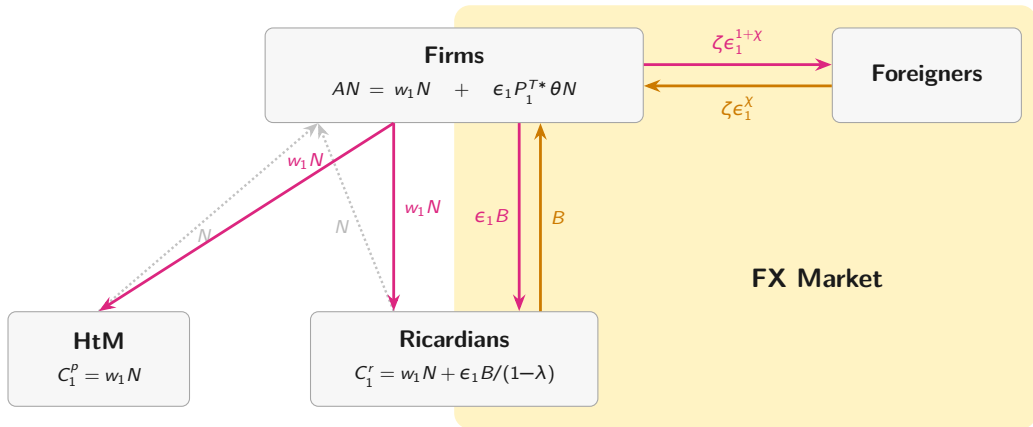
Optimal labor wedge:

$$v'(N_t) = \bar{u}'_t \left(w_t - \frac{\epsilon_t P_t^{T*} \theta}{\chi} \right) + \lambda [u'(C_t^p) - u'(C_t^r)] NX_t \frac{d\epsilon_t}{dN_t} \Big|_{NX_t}$$

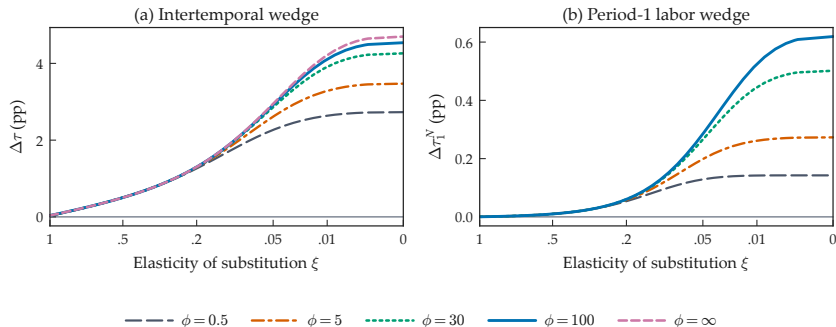
$$\tau_t^N = \frac{1}{\chi} \frac{\epsilon_t P_t^{T*} \theta}{w_t} - \frac{\lambda [u'(C_t^p) - u'(C_t^r)] NX_t \frac{d\epsilon_t}{dN_t} \Big|_{NX_t}}{\bar{u}'_t w_t}$$

Flow of Goods: The Foreign Exchange Market

- Suppose $P_1^{T*} > P_2^{T*}$ so that Ricardians are net borrowers at $t = 1$ with $B \equiv -F^*$.
- Ricardians and foreigners are net suppliers of the **foreign good** and demand the **domestic good**; firms are net demanders of the foreign good and supply the domestic good in exchange.

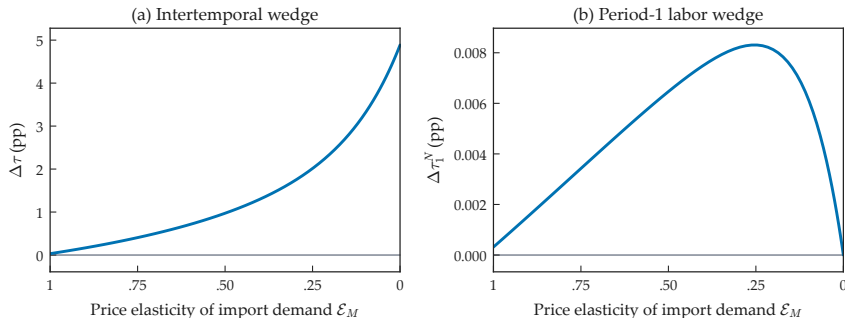


Comparative Statics: CES Production



- Generalize production to CES:
$$Y_t = A_t \left[(1 - \alpha) N_t^{\frac{\xi-1}{\xi}} + \alpha \left(\frac{M_t}{\theta} \right)^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}}.$$
- Comparative statics with respect to the price elasticity of import demand. Plotted case: $P_1^{T*} > P_2^{T*}$, optimal wedge differences between $\lambda > 0$ and $\lambda = 0$ economies. Increasing as $\xi \rightarrow 0$ and $\phi \rightarrow \infty$: **essential imports strengthen both policy motives** under heterogeneity.

Comparative Statics: Imports as Final Goods



- Imported good as a final good with Stone-Geary preferences: $u(C, M) = \frac{[C^{1-\omega}(M-\underline{M})^\omega]^{1-\sigma}}{1-\sigma}$, $M \geq \underline{M}$.
- Comparative statics with respect to the price elasticity of import demand. Plotted case: $P_1^{T*} > P_2^{T*}$. Optimal wedge differences between $\lambda > 0$ and $\lambda = 0$ economies.
- **FX policy's role increasing with essentiality:** $\Delta\tau$ increases as import demand becomes inelastic.
- $\Delta\tau_1^N$ small, weaker role for monetary policy. Perfectly inelastic demand: heterogeneity adds no role.

- The competitive labor packer combines differentiated tasks into final labor:

$$N_t = \left(\int_0^1 N_{k,t}^{\frac{\varepsilon_w - 1}{\varepsilon_w}} dk \right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}}, \quad \varepsilon_w > 1$$

- It solves

$$\min_{\{N_{k,t}\}_{k \in [0,1]}} \int_0^1 W_{k,t} N_{k,t} dk \quad \text{s.t.} \quad \left(\int_0^1 N_{k,t}^{\frac{\varepsilon_w - 1}{\varepsilon_w}} dk \right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}} \geq N_t$$

- Cost minimization yields

$$N_{k,t} = \left(\frac{W_{k,t}}{W_t} \right)^{-\varepsilon_w} N_t, \quad W_t = \left(\int_0^1 W_{k,t}^{1-\varepsilon_w} dk \right)^{\frac{1}{1-\varepsilon_w}}$$

Unions' Wage-Setting Problem and the WPC

- Taking the task-demand schedule as given, union k chooses its nominal wage sequence to solve

$$\max_{\{W_{k,t+s}\}_{s \geq 0}} \sum_{s \geq 0} \beta^s \left[\lambda u(C_{t+s}^p) + (1-\lambda)u(C_{t+s}^r) - v(N_{t+s}) - \frac{\psi_{nr}}{2} \left(\frac{W_{k,t+s}}{W_{k,t+s-1}} - 1 \right)^2 \right]$$

subject to

$$N_{k,t+s} = \left(\frac{W_{k,t+s}}{W_{t+s}} \right)^{-\varepsilon_w} N_{t+s}.$$

- Each union is infinitesimal and takes W_{t+s} and N_{t+s} as given.
- In a symmetric equilibrium, $W_{k,t} = W_t$, $N_{k,t} = N_t$ for every k . Wage Phillips curve:

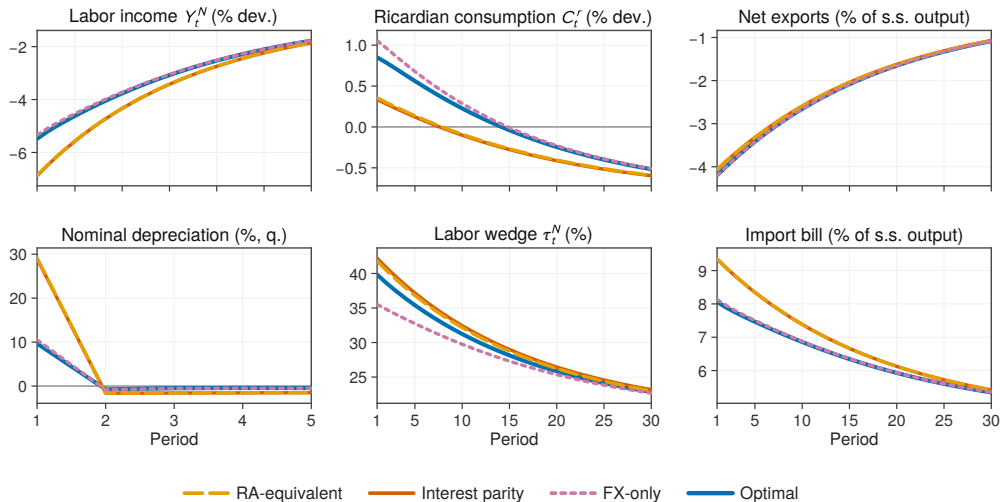
$$v'(N_t) = (1 - \tau_t^N) w_t \bar{u}'_t,$$

$$1 - \tau_t^N = \frac{1}{\mu_w} + \frac{\left(\frac{w_t}{w_{t-1}} \Pi_t - 1 \right) \frac{w_t}{w_{t-1}} \Pi_t - \beta \left(\frac{w_{t+1}}{w_t} \Pi_{t+1} - 1 \right) \frac{w_{t+1}}{w_t} \Pi_{t+1}}{\frac{\varepsilon_w}{\psi_{nr}} w_t N_t \bar{u}'_t}$$

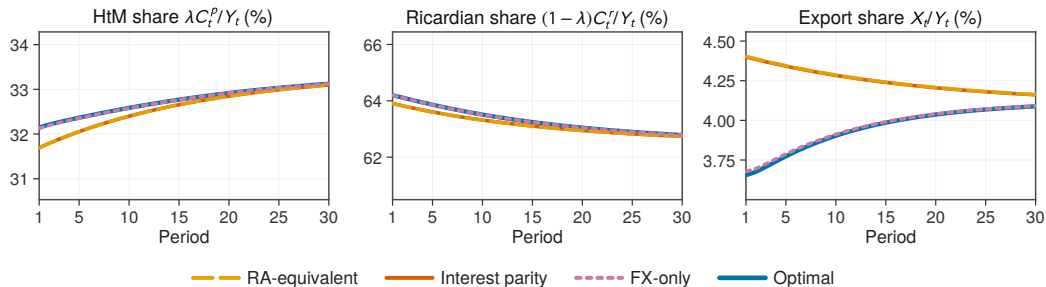
Quarterly calibration for Japan

	Parameter	Value	Target or source
R^*	Gross world interest rate	1.0021	Net exports of -0.5% of GDP
β	Discount factor	0.9979	$\beta R^* = 1$
σ	Inverse intertemporal elasticity	4.0	Intertemporal elasticity of 0.25
η	Labor-disutility scale	0.9261	Steady-state employment normalization
ϕ	Inverse Frisch curvature	29.36	Output falls 1% on impact after a 100% import-price rise in the RA economy
λ	HtM household share	0.35	Kaplan, Violante, and Weidner (2014) ballpark for the US
ζ	Export demand scale	1.9758	Fossil-fuel import bill of 4%
χ	Marshall-Lerner elasticity	0.25	Modest short-run response: Rose (1991); Bahmani-Oskooee and Ratha (2004)
ρ_{PT^*}	Import-price persistence	0.945	AR(1) on detrended IMF energy price index (IMF, via FRED)
Γ	Intermediation friction	9.0	Fanelli and Straub (2021)
ε_w	Wage substitution elasticity	6.129	Core CPI inflation of 0.097% per quarter (2014Q4–2019Q4) in RA steady state
ψ_{nr}	Nominal wage rigidity	10598.1	Core CPI inflation of 0.411% per quarter (2020Q4–2022Q3) in RA 2022 spike
$B_{ss}^{G^*}$	Public foreign assets	60.0	Net foreign assets of 60% of annual GDP

Transitory Spike: Additional Outcomes

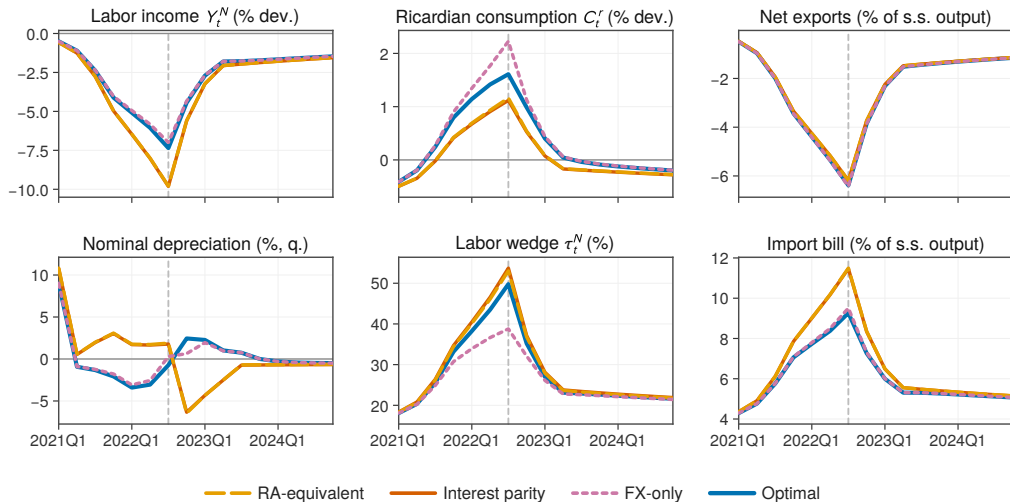


Transitory Spike: Output Shares

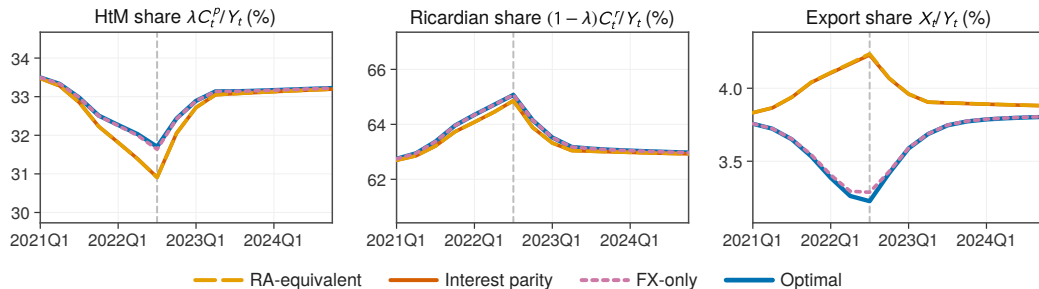


- During the spike, optimal policy retains more output for domestic consumption and reduces the export share. HtM share goes up due to indirect income effects.

2022 Spike: Additional Outcomes



2022 Spike: Output Shares



- Around the peak of the crisis, optimal policy lowers the export share and retains more output for domestic consumption. HtM share goes up due to indirect income effects.

Heterogeneous Agent Framework

- Unit mass of households, autonomous consumption requirement z_t^i .
- $z_{t+1}^i \sim Z(\cdot | z_t^i)$, invariant π_z , fixed aggregate mean $\int z d\pi_z = \bar{z}$: no aggregate risk.
- Inflation-indexed domestic asset market. Common labor supply N_t as before.
- Household problem:

$$\max_{\{c_t^i, b_{t+1}^i\}} \mathbb{E}_1 \sum_{t=1}^{\infty} \tilde{\beta}^{t-1} [u(c_t^i) - v(N_t)]$$

$$c_t^i + b_{t+1}^i = R_t b_t^i + w_t N_t - z_t^i + (R_t - 1)\underline{b}, \quad b_{t+1}^i \geq 0$$

- Optimality conditions:

$$u'(c_t^i) \geq \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i), \quad b_{t+1}^i [u'(c_t^i) - \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i)] = 0$$

- FX policy with no transfers: public agent subject to period budgets and no-Ponzi conditions.

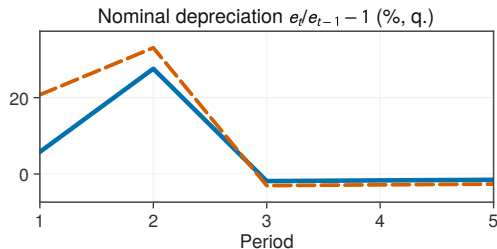
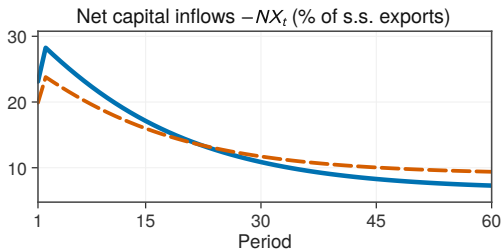
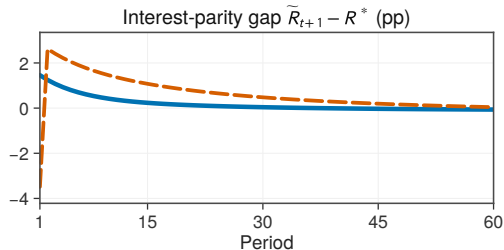
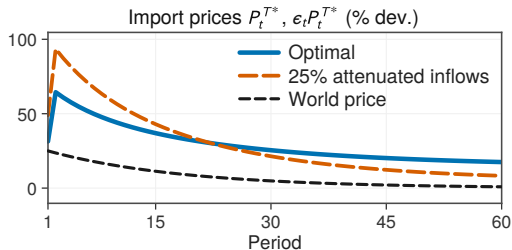
$$B_{t+1}^G + \epsilon_t B_{t+1}^{G*} = R_t B_t^G + \epsilon_t R^* B_t^{G*}$$

Heterogeneous Agents: Ramsey Problem under Timeless Commitment

$$\begin{aligned}
 & \max_{\{c_t^i, b_{t+1}^i, N_t, \epsilon_t, R_{t+1}\}_{t \geq 1}} \mathbb{E}_1 \sum_{t=1}^{\infty} \beta^{t-1} \left[\int u(c_t^i) di - v(N_t) \right] - \frac{\tilde{\beta}}{\beta} R_1 \int \Theta_0^i u'(c_1^i) di - \frac{\psi_0}{\beta \Gamma R^*} \frac{R_1 \epsilon_0}{\epsilon_1} \\
 & c_t^i + b_{t+1}^i = R_t b_t^i + (A_t - \epsilon_t P_t^{T*} \theta) N_t - z_t^i + (R_t - 1) \underline{b}, \quad b_{t+1}^i \geq 0 \\
 & u'(c_t^i) \geq \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i), \quad b_{t+1}^i [u'(c_t^i) - \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i)] = 0 \\
 & \int c_t^i di + \bar{z} + \zeta \epsilon_t^{1+\chi} = A_t N_t \\
 & \sum_{t=1}^{\infty} \frac{P_t^{T*} \theta N_t + \frac{1}{\Gamma} \left(1 - \frac{R_{t+1}}{R^*} \frac{\epsilon_t}{\epsilon_{t+1}} \right)^2 - \zeta \epsilon_t^\chi}{(R^*)^{t-1}} = R^* B_1^{G*} + R_1 \frac{\epsilon_0 F_1^*}{\epsilon_1}
 \end{aligned}$$

- Initial states: $\{b_1^i, z_1^i, \Theta_0^i\}_i, R_1, \epsilon_0, F_1^*, B_1^{G*}, \psi_0$
- With no lump-sum transfers: household budget constraints and transversality are implementability conditions for the Ramsey planner.
- Timeless commitment: past promises encoded in Lagrange multipliers (Marcet and Marimon, 2019).

Heterogeneous Agents: Additional Policy Outcomes



- Counterfactual policy: 25% attenuated net capital inflows on impact, employment fixed at optimal.